1.

$$\rho(x) = \begin{cases} -qN_A, & -x_p \le x \le 0 \\ +qN_D, & 0 \le x \le x_n \\ 0, & |x| > \text{depletion region} \end{cases}$$

Here, q is the electronic charge, N_A is the acceptor concentration(m^{-3}) in the p-region, and N_D is the donor concentration(m^{-3}) in the n-region. The depletion region extends from $-x_p$ to x_n. At equilibrium, mobile carriers are depleted in the region $-x_p \le x \le x_n$, leaving behind fixed ionized dopants that give rise to the space charge density $\rho(x)$.

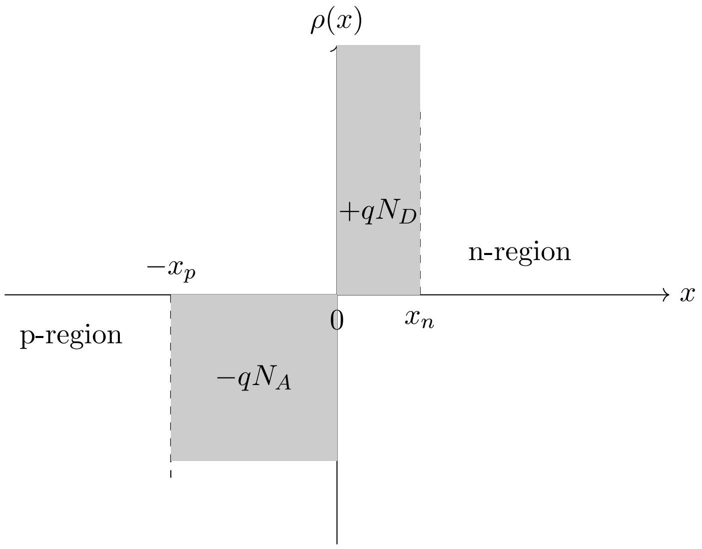


Figure 1: Charge distribution

(a) Identify what the free charges are in this case and understand what the term "free" truly refers to.

(b) Find the potential drop across the junction.

(c) Plot the $\vec{E}(x)$ Vs x.

(d) Plot the $V(x)$ Vs x.

2. Consider the two scalar potential functions

$$V_1(x, y) = x \text{ and } V_2(x, y) = x + e^{-y} \sin x.$$

(a) Show that both V_1 and V_2 satisfy Laplace's equation.

(b) Verify that both potentials satisfy the boundary condition $V = 0$ on the boundary $x = 0$.

(c) Are V_1 and V_2 identical everywhere in space? Justify your answer.

(d) Does the uniqueness theorem apply to this problem? Explain clearly which assumptions of the theorem are or are not satisfied.

3. Two concentric conducting spheres of radii $a = 5$ mm and $b = 20$ mm are separated by air. The inner sphere is maintained at a potential of 100 V, while the outer sphere is grounded.

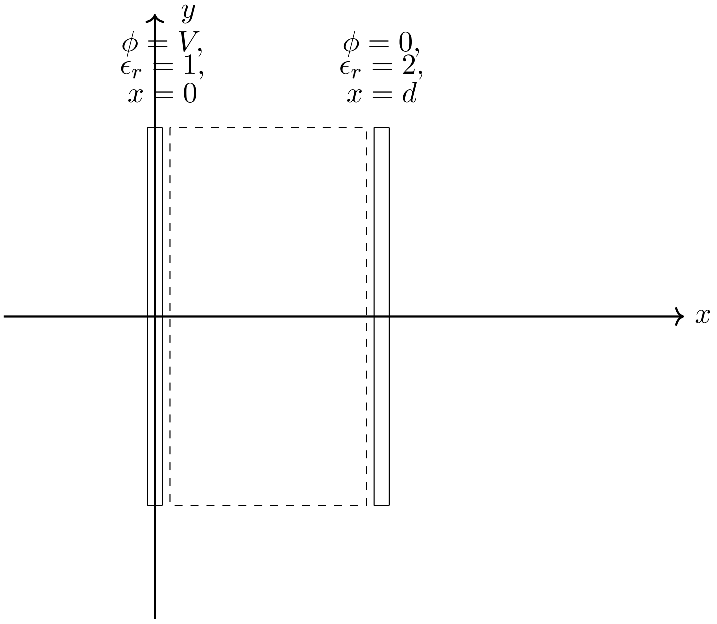
   (a) Assuming electrostatic equilibrium, determine the electric potential $V(r)$ in the region $a \leq r \leq b$ by solving Laplace's equation, and verify the result by comparison with the electric field obtained from Gauss's law.

   (b) Obtain an expression for the radial electric field $E_r(r)$ in the region between the spheres and determine the location and value of the maximum electric field.

   (c) Find the radius at which the electric potential is 20 V.

   (d) Plot the electric potential and electric field in a two–dimensional circular cross–section of the region between the spheres, clearly indicating the equipotential surfaces and field direction.

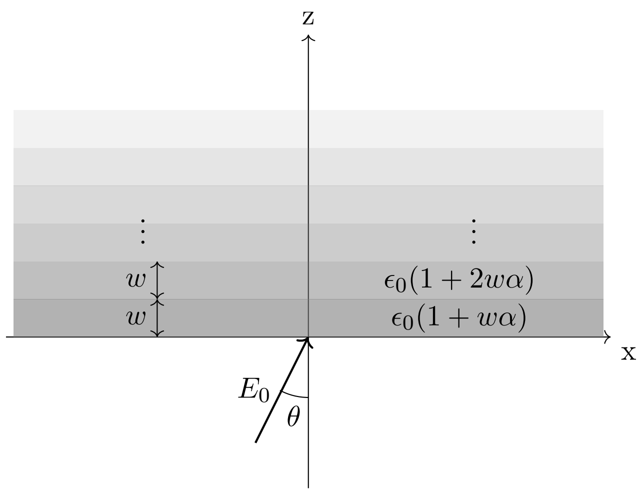
4. The space between the plates of a parallel-plate capacitor is filled with dielectric material whose dielectric constant varies linearly from 1 at the leftmost plate (x = 0) to 2 at the rightmost plate (x = d). The capacitor is connected to a battery of voltage V. Find all the bound charge, and check that the total is zero.



5. Consider a two-dimensional system in the x-z plane where the region $z > 0$ is filled with horizontal dielectric slabs of width w. The i-th slab has the permittivity

$$\epsilon_i = \epsilon_0(1 + \alpha i w), \quad \alpha > 0$$

An electric field enters the region from $z < 0$, making an angle θ with the z-axis.

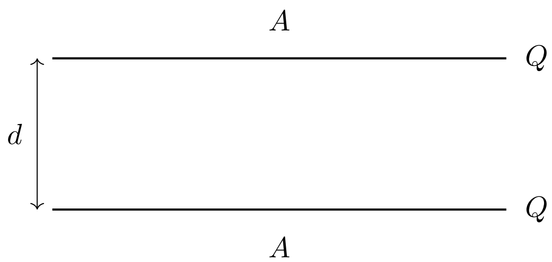


(a) Using the standard boundary conditions at each interface between layers, explain qualitatively how the electric field bends at each layer boundary.

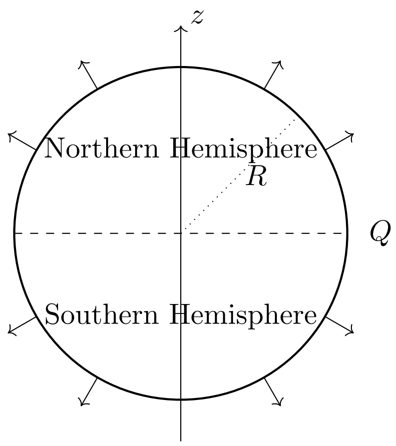
(b) Sketch a field line passing through several layers, showing how it changes direction at each interface.

(c) Consider the case where $w \to 0$. Explain how the field trajectory changes in this limit and sketch a field line.

6. (a) Two large parallel metal plates, each of area A, are held a small distance d apart. Each plate is given a charge Q.Assuming edge effects are negligible, find the **electrostatic pressure** exerted on the plates and hence determine the force acting on each plate.



(b) A conducting metal sphere of radius R carries a total charge Q. What is the force of repulsion between the **northern hemisphere** and the **southern hemisphere** of the sphere?



7. A coaxial cable consists of an inner conducting cylinder of radius a and an outer conducting cylinder of radius b. The space between them is filled with two dielectric layers:

   - Region 1: $a < r < c$ with permittivity ϵ_1
   - Region 2: $c < r < b$ with permittivity ϵ_2

   The inner conductor carries charge per unit length $+\lambda$, and the outer conductor carries $-\lambda$.

   (a) Find the electric field in both dielectric regions.
   (b) Determine the potential difference between the conductors.
   (c) Obtain the capacitance per unit length of this structure.
   (d) Find the electrostatic energy stored per unit length.
   (e) Determine the bound surface charge density at the interface $r = c$.

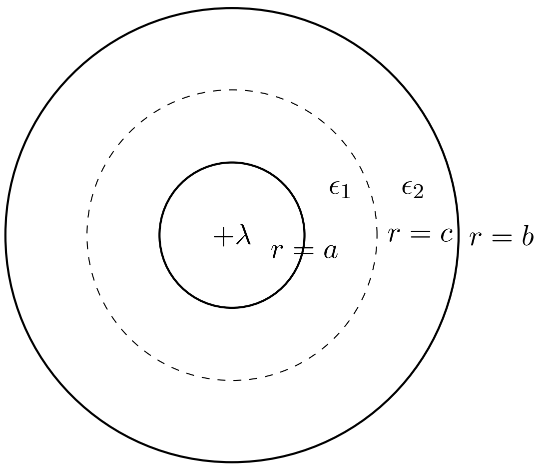


Figure 2: Coaxial capacitor with two dielectric layers